

SECTION 2

LIMITATIONS

TABLE OF CONTENTS

	<u>Page</u>
2.1 GENERAL	2 - 1
2.2 KINDS OF OPERATION	2 - 1
2.3 BASIS OF CERTIFICATION	2 - 1
2.4 OCCUPANTS	2 - 1
2.4.1 Flight Crew	2 - 1
2.5 FLIGHT WITH OPTIONAL EQUIPMENT INSTALLED	2 - 2
2.6 MASS AND LOAD LIMITATIONS	2 - 2
2.6.1 Maximum Gross Mass	2 - 2
2.6.2 Minimum Gross Mass	2 - 2
2.6.3 Loading	2 - 2
2.6.4 Cargo Securing Points	2 - 2
2.7 CENTER OF GRAVITY LIMITATIONS	2 - 3
2.7.1 Longitudinal Center of Gravity	2 - 3
2.7.2 Lateral Center of Gravity	2 - 3
2.8 AIRSPPEED LIMITATIONS	2 - 4
2.8.1 Forward Flight	2 - 4
2.8.2 Sideward Flight	2 - 5
2.8.3 Rearward Flight	2 - 5
2.9 ALTITUDE LIMITATIONS	2 - 5
2.10 ENVIRONMENTAL OPERATING CONDITIONS	2 - 7
2.10.1 Altitude/Ambient Air Temperature Limitations	2 - 7
2.10.2 Engine Anti-icing	2 - 7
2.10.3 Pitot Heating	2 - 7
2.10.4 Icing Conditions	2 - 7

	<u>Page</u>
2.10.5 Snow Conditions	2 - 7
2.10.6 Battery Temperature Limitations	2 - 7
2.11 ROTOR RPM LIMITATIONS	2 - 8
2.12 ENGINE AND TRANSMISSION POWER LIMITATIONS	2 - 10
2.13 OTHER ENGINE AND TRANSMISSION LIMITATIONS	2 - 11
2.13.1 Engine Oil Pressure	2 - 11
2.13.2 Engine Oil Temperature	2 - 11
2.13.3 Engine Starter	2 - 12
2.13.4 Generator	2 - 12
2.13.5 Ground Power Starts	2 - 12
2.13.6 Main Transmission Oil Pressure	2 - 12
2.13.7 Main Transmission Oil Temperature	2 - 13
2.14 FUEL LIMITATIONS	2 - 13
2.14.1 Fuel Pressure	2 - 13
2.14.2 Fuel Specifications	2 - 13
2.14.3 Fuel Specifications - Cold Weather	2 - 14
2.14.4 Fuel Additives	2 - 14
2.14.5 Fuel Temperature	2 - 15
2.14.6 Fuel Quantities	2 - 15
2.15 OIL LIMITATIONS	2 - 16
2.15.1 Oil Specifications	2 - 16
2.15.2 Oil Quantities	2 - 16
2.16 HYDRAULIC SYSTEM LIMITATIONS	2 - 16
2.16.1 Operational Hydraulic System	2 - 16
2.16.2 Hydraulic Fluid	2 - 16
2.17 OPERATIONAL LIMITATIONS	2 - 16
2.17.1 Rotor Starting and Stopping in Wind	2 - 16
2.17.2 Hover Turn	2 - 17
2.17.3 Main Rotor Mast Moment	2 - 17
2.17.4 Slope Operations	2 - 17
2.17.5 Prohibited Flight Maneuvers	2 - 18

	<u>Page</u>
2.17.6 Mass-altitude-temperature	2 - 18
2.17.7 Extreme Low Temperature Operation	2 - 18
2.17.8 Engine Inlet Deflector Shield	2 - 18
2.18 INSTRUMENT MARKINGS	2 - 19
2.18.1 Airspeed Indicator	2 - 19
2.18.2 Engine RPM (N_1) Indicator	2 - 19
2.18.3 Triple Tachometer	2 - 20
2.18.4 Turbine Outlet Temperature (TOT) Indicator	2 - 20
2.18.5 Triple Oil Pressure Indicator	2 - 21
2.18.6 Triple Oil Temperature Indicator	2 - 21
2.18.7 Dual Torque Indicator	2 - 22
2.18.8 Fuel Pressure Indicator	2 - 22
2.19 PLACARDS AND DECALS	2 - 23

LIST OF FIGURES

	<u>Page</u>
Fig. 2-1 Allowable CG Envelope (Longitudinal)	2 - 3
Fig. 2-2 Never-exceed Speed (V_{NE})	2 - 4
Fig. 2-3 Altitude/Temperature Limit Diagram	2 - 6
Fig. 2-4 Temperature Limits for Approved Primary Fuels	2 - 15

